

Question	Marking Guidance	Mark	Comments
08.1	Correct answer of 7 (days) = 2 marks;; Incorrect answer but shows 7 to incorrect magnitude e.g. 700 (ignore preceding/subsequent zeros and position of decimal point i.e. did not divide by 100 or incorrect use of order of magnitude) = 1 mark OR Incorrect answer of 168 (incorrectly multiplied by 24) = 1 mark OR Incorrect answer shows 9 ignoring subsequent numbers (multiplied rather than divided by 1.14) = 1 mark OR Incorrect but shows 6.93 (incorrect rounding of 7.98) = 1 mark OR Incorrect but shows 7.98 or 8 (did not divide by 1.14) = 1 mark;	2 (2 x AO2)	
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08.2	1. (<i>E. coli</i> / prokaryotes/bacteria) do not have introns /non-coding (DNA) OR (<i>E. coli</i> / prokaryotes/bacteria) only have exons / coding (DNA) OR Eukaryotes contain introns / non-coding (DNA); 2. Eukaryotes contain (more) regulatory genes;	2 (2 x AO1)	1 and 2 Accept 'It' refers to <i>E. coli</i> 2. Accept prokaryotes have no/fewer regulatory genes

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08.3	1. (Identification/isolation of) <u>antigens</u> ; 2. (Antigens/proteins used to develop) vaccines/antitoxins/antiserums OR (Antigens/proteins used) to develop (monoclonal) antibodies;	2 (2 x AO1)	2. Accept descriptions of (active) immunity 2. Ignore antibiotics

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08.4	Mark points 1 and 2 are required for maximum marks. 1. Two repeats decreases (risk of) breast cancer; 2. Three and four repeats increases (risk of) breast cancer; 3. No results for more than 4 repeats OR No results for 1 repeat OR Few repeats (studied); 4. Other factors affect (risk of) breast cancer OR Named factor (e.g. age) affects (risk of) breast cancer; 5. Breast cancers not all same (type); 6. Sample size not known;	4 max (4 x AO3)	Ignore length of investigation and use of a statistical test 2. Accept 'more than two repeats' 3. Accept only 2,3 and 4 repeats Ignore increase in VNTRs repeats increase risk of breast cancer
